UNIT 10 · TOPIC 8

Interpretive Products — Maps, Charts and Application Maps

Syllabus topic: Interpretive products: maps, charts and application maps

Maps for decision-making

A MAP ties an attribute to a location.

Map	Source
Point/grid soil-sampling maps	Georeferenced soil samples
YIELD map	Combine yield monitor + GPS (Topic 5)
Soil EC and nutrient maps	EC sensors, laboratory analysis
Elevation maps	GPS/RTK surveys
NDVI / vegetation maps	Remote sensing (Topic 6)
APPLICATION (prescription) map	Tells the variable-rate controller how much to apply at each point
As-applied map	Logged by the controller during application

Spatial analysis — turning points into decisions

1. **INTERPOLATION** (kriging/IDW) builds continuous surfaces from samples.
2. **OVERLAY** combines the layers.
3. **MANAGEMENT ZONES** group areas of similar characteristics for practical differential treatment.
4. The outputs — application maps and as-applied maps — close the PA loop and allow season-to-season learning.

Practice MCQs

Q5. A map produced from a combine sensor plus GPS is a:

(a) Soil pH map (b) Yield map
(c) Road map (d) Contour of rainfall

Q6. Grouping areas of similar soil/yield for differential treatment gives:

(a) Management zones (b) Field boundaries
(c) Grid north (d) Datum

Q68. Delineating areas of similar soil/yield for differential management creates:

(a) Management zones (b) Field roads
(c) Contours (d) Grids of DOP

Answers

Q5 — (b) Yield map
Q6 — (a) Management zones
Q68 — (a) Management zones

Sources: Srinivasan — Handbook of Precision Agriculture; NIIT/ICAR precision-agriculture modules; NovAtel — Introduction to GNSS; ISRO/PIB NavIC data (as given in the source notes, July 2026; operational status updated October 2026); NPTEL precision-agriculture lectures. Srinivasan — Handbook of Precision Agriculture; ICAR/NIIT PA modules; Lillesand & Kiefer — Remote Sensing and Image Interpretation; ISO 11783 (ISOBUS); remote-sensing and NDVI references (as given in the source notes, July 2026); NPTEL precision-agriculture lectures.
